AMATH/PMATH 331: Applied Real Analysis Fall 2026

Midterm test; Solutions

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[1] Let $\mathbb{Z} \subset \mathbb{R}$ be the set of integers.

[a] Prove that $\mathbb{Z}$ is closed.
Hint: Show that $\mathbb{R} \setminus \mathbb{Z}$ is open using the Archimedean property.

[b] We showed in class that, in general, an infinite union of closed sets need not be closed. Given an explicit example of infinitely many *pairwise disjoint* nonempty closed sets whose union is closed. Be sure to completely justify your answer.

Solution.

[a] Let $x \in \mathbb{R} \setminus \mathbb{Z}$. By the Archimedean property, $\exists\, n \in \mathbb{Z}$ such that $n < x < n+1$. (Note that $x \neq n$ because $x \notin \mathbb{Z}$.) Let $\delta = \min\{x - n, (n+1) - x\} > 0$. Then $B_\delta(x) \subseteq \mathbb{R} \setminus \mathbb{Z}$, so $\mathbb{R} \setminus \mathbb{Z}$ is open, and hence $\mathbb{Z}$ is closed.

[b] There are many possible answers, but the hint for an easy example is from part [a]. For $n \in \mathbb{Z}$, let $F_n = \{n\}$. Then the F_n are nonempty, closed, and pairwise disjoint. And $\cup_{n\in\mathbb{Z}} F_n = \mathbb{Z}$ is closed.

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[2] Let $E \subseteq \mathbb{R}^n$. We say that E is *dense* iff $\overline{E} = \mathbb{R}^n$, where $\overline{E}$ is the closure of E.

[a] Prove that E is dense in $\mathbb{R}^n$ iff: given any $\mathbf{x} \in \mathbb{R}^n$ and any $\varepsilon > 0$, there exists $\mathbf{p} \in E$ such that $\|\mathbf{x} - \mathbf{p}\| < \varepsilon$.

[b] Consider the rational numbers $\mathbb{Q} \subset \mathbb{R}$. Explain why $\mathbb{Q}$ is dense in $\mathbb{R}$.

Solution.

[a] Recall that we defined the closure by

$$\overline{E} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap E \neq \varnothing \text{ for all } \varepsilon > 0\}.$$

Thus we have

$$\begin{aligned} \overline{E} = \mathbb{R}^n &\iff \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap E \neq \varnothing \text{ for all } \varepsilon > 0\} = \mathbb{R}^n \\ &\iff \text{for all } \mathbf{x} \in \mathbb{R}^n \text{ and any } \varepsilon > 0,\ \exists\, p \in E \text{ such that } \mathbf{p} \in B_\varepsilon(\mathbf{x}) \\ &\iff \text{for all } \mathbf{x} \in \mathbb{R}^n \text{ and any } \varepsilon > 0,\ \exists\, p \in E \text{ such that } \|\mathbf{x} - \mathbf{p}\| < \varepsilon. \end{aligned}$$

[b] Let $x \in \mathbb{R}$ and $\varepsilon > 0$. We proved in class there exists $q \in \mathbb{Q}$ such that $x - \varepsilon < q < x + \varepsilon$. Hence $B_\varepsilon(x) \cap \mathbb{Q} \neq \varnothing$. So $x \in \overline{\mathbb{Q}}$.

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[3] Let $(\mathbf{x}_k)$ be a sequence in $\mathbb{R}^n$.

[a] Define what it means for $(\mathbf{x}_k)$ to be *Cauchy*.

[b] What is the relationship between Cauchy sequences and convergent sequences?

[c] Suppose $(\mathbf{x}_k)_{k\in\mathbb{N}}$ is Cauchy, and that it has a *subsequence* $(\mathbf{x}_{f(k)})_{k\in\mathbb{N}}$ which converges to $\mathbf{p} \in \mathbb{R}^n$. Prove that the original sequence $(\mathbf{x}_k)_{k\in\mathbb{N}}$ also converges to $\mathbf{p}$.
[Be sure to state clearly any results you use to prove this.]

Solution.

[a] $(\mathbf{x}_k)$ is Cauchy iff: for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $k, \ell \geq N$ implies $\|\mathbf{x}_k - \mathbf{x}_\ell\| < \varepsilon$.

[b] We proved in class that a sequence is Cauchy if and only if it converges.

[c] Suppose $(\mathbf{x}_k)$ is Cauchy, so it converges to some limit $\mathbf{q}$. But we proved in class that if a sequence converges to a limit, then *any* subsequence of it also converges, and to the same limit. So $\mathbf{q} = \mathbf{p}$.

[4] Let $\mathbb{R}^{2\times 2}$ be the space of 2×2 real matrices:

$$\mathbb{R}^{2\times 2} = \left\{ \begin{pmatrix} x & y \\ z & w \end{pmatrix} : x, y, z, w \in \mathbb{R} \right\}.$$

[a] Consider the determinant map $\det\colon \mathbb{R}^{2\times 2} \to \mathbb{R}$ that takes $A \in \mathbb{R}^{2\times 2}$ to $\det A \in \mathbb{R}$. Explain why det is a *continuous map*.

[b] Prove that the subset $\mathcal{I} \subset \mathbb{R}^{2\times 2}$ of *invertible* 2×2 matrices is an *open* subset.
Hint: You can use the fact that $\{t \in \mathbb{R} : t \neq 0\}$ is an open subset of $\mathbb{R}$.

[c] Is $\mathcal{I}$ a *bounded* subset of $\mathbb{R}^{2\times 2}$? Justify your answer.

[d] Is $\mathcal{I}$ a *compact* subset of $\mathbb{R}^{2\times 2}$? Justify your answer?

Solution.

[a] We have

$$\det\left(\begin{pmatrix} x & y \\ z & w \end{pmatrix} \right) = xw - yz$$

is a polynomial in x, y, z, w, and hence continuous everywhere.

[b] A matrix A is invertible if and only if $\det A \neq 0$. Thus

$$\mathcal{I} = \det^{-1}(\{t \in \mathbb{R} : t \neq 0\})$$

is the inverse image of an open set. Since det is continuous on $\mathbb{R}^{2\times 2}$, it follows that $\mathcal{I}$ is open.

[c] No, $\mathcal{I}$ is not bounded. For example, for any $M > 0$, take

$$A = \begin{pmatrix} M & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $A \in \mathcal{I}$, and $\|A\| = \sqrt{M^2 + 1}$.

[d] No, $\mathcal{I}$ is not compact. Any compact set must be closed and bounded, but $\mathcal{I}$ is not bounded.